

UTA025-PROJECT REPORT

SmartSwipe

Intelligent Expense Optimization System



Submitted By :-

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Executive Summary

SmartSwipe is an AI-powered personal finance management platform designed to revolutionize the way individuals interact with their credit cards and reward systems. As the Indian credit card market expands rapidly—with over 98 million cards in circulation as of 2025—most cardholders fail to leverage the full reward potential embedded in their spending patterns. SmartSwipe addresses this critical gap through a real-time, AI-driven recommendation engine that analyzes transaction contexts and suggests the optimal card for every purchase.

The system integrates a multi-layered architecture comprising a React Native mobile frontend, a Django REST Framework backend, a PostgreSQL-based relational database, and a hybrid AI engine utilizing collaborative filtering and rule-based inference. By aggregating data from multiple credit card profiles, spending categories, merchant networks, and promotional offers, SmartSwipe provides users with personalized, actionable insights.

Our customer validation survey of 120 respondents confirms strong market demand: 78% of multi-card holders reported confusion in selecting the right card, and 84% expressed willingness to adopt an AI-based optimization tool. The freemium business model projects a revenue trajectory from Rs. 2.4 lakhs in Year 1 to Rs. 28.7 lakhs by Year 3, driven by premium subscriptions and affiliate commissions from banking partners.

SmartSwipe stands apart from existing solutions such as CRED and ET Money by offering real-time transactional optimization—not just post-purchase analytics. This project report documents the complete lifecycle of SmartSwipe from opportunity identification and market validation through to financial modeling and implementation planning.

Chapter 1: Opportunity Analysis

1.1 Opportunity Identification

India's digital payment landscape has undergone a dramatic transformation over the past five years. Credit card issuances grew at a compound annual growth rate (CAGR) of 22% between 2020 and 2025, and the Reserve Bank of India (RBI) reported a 34% year-on-year increase in credit card transactions in FY 2024-25. Yet despite this exponential growth, research indicates that approximately 67% of credit card users in India do not optimize their rewards—either selecting the wrong card for a given transaction category or completely ignoring reward structures.

The root cause is information asymmetry: each credit card carries a complex, ever-changing array of reward rates, cashback tiers, partner merchant offers, bonus categories, and redemption restrictions. Managing three or more credit cards effectively requires continuous monitoring of multiple bank portals, periodic rule updates, and transaction-level decision-making that most users find cognitively burdensome.

The identified opportunity is the creation of an intelligent middleware layer that sits between the user and their card ecosystem—one that understands spending context in real time and recommends the highest-reward option at the point of transaction. This is the precise problem SmartSwipe is engineered to solve.

1.2 Proposed Solution

SmartSwipe is a mobile-first, AI-powered recommendation platform that aggregates a user's credit card portfolio and spending history to deliver real-time, context-sensitive card recommendations. The core solution comprises four integrated components:

- **Reward Intelligence Engine:** A hybrid AI model trained on card-specific reward structures that evaluates every transaction against all eligible cards and ranks them by net reward value.
 - **Unified Dashboard:** A centralized interface displaying all linked cards, cumulative reward balances, expiry alerts, and redemption opportunities.
 - **Spending Analytics Module:** Category-wise spending breakdowns with trend analysis, budget forecasting, and personalized saving suggestions.
 - **Smart Notification System:** Push notifications triggered at the moment of transaction decision, ensuring optimal card selection without cognitive effort from the user.
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1.3 Uniqueness of Solution

SmartSwipe's competitive differentiation rests on three pillars that no existing consumer application addresses simultaneously:

Feature	SmartSwipe	CRED	ET Money
Real-time card recommendation	Yes	No	No
Multi-card optimization	Yes	Limited	No
AI personalization	Yes	Partial	Partial
Reward expiry alerts	Yes	Yes	No
Spending analytics	Yes	Basic	Yes
Free tier available	Yes	Yes	Yes

The key differentiator is the pre-transaction, real-time recommendation—a capability not available in any major Indian fintech application as of 2026. This transforms SmartSwipe from a passive analytics tool into an active financial co-pilot.

1.4 User Identification and Market Segmentation

SmartSwipe targets five clearly defined customer segments, each with a distinct value proposition:

Segment	Description	Key Benefit Delivered
Multi-Card Holders	Individuals with 3+ credit cards across banks	Eliminates card-selection confusion; maximizes cross-card rewards
Young Professionals	25–35 age group with growing disposable income	Builds healthy credit behavior; maximizes early savings
Online Shoppers	Frequent e-commerce users on Flipkart, Amazon, Myntra	Identifies highest cashback card per merchant category
Frequent Travelers	Business travelers and leisure travelers using travel cards	Optimizes lounge access, air miles, and travel insurance
Finance-Aware Users	Premium users seeking deep financial analytics	Advanced insights, budget forecasting, custom reward goals

The total addressable market (TAM) is estimated at 98 million credit card users in India; the serviceable obtainable market (SOM) for the first two years is targeted at 250,000 active users in Tier-1 and Tier-2 cities.

1.5 Product Features and Adoption Strategy

Core Features

- Real-Time Card Recommendation Engine with merchant category mapping
- Unified Multi-Card Dashboard with reward balance aggregation
- AI-Powered Spending Categorization (dining, travel, groceries, fuel, entertainment)
- Reward Expiry Tracker with automated alerts 30/15/7 days in advance
- Monthly Spend Reports with visual analytics
- Secure Bank API Integration (via OAuth 2.0 / Account Aggregator framework)
- Gamified Savings Goals and Reward Milestones

Adoption Strategy

Phase 1 (Months 1–3): Beta launch targeting 500 early adopters via college campuses and LinkedIn communities. Focus on product-market fit validation.

Phase 2 (Months 4–9): Public launch on Play Store and App Store. Influencer marketing via personal finance YouTube channels and Instagram. Referral program offering 1-month premium free per successful referral.

Phase 3 (Months 10–18): Bank partnership integrations for co-branded features. Corporate tie-ups for employee financial wellness programs. Expansion to Tier-2 cities.

Chapter 2: Customer Validation Survey

2.1 Survey Overview

A structured customer validation survey was conducted between February 15 and March 20, 2026, targeting credit card users across major Indian cities including Delhi, Mumbai, Bangalore, Pune, Hyderabad, and Chandigarh. The survey was administered through Google Forms and distributed via WhatsApp groups, LinkedIn, college alumni networks, and direct outreach. A total of 120 valid responses were recorded, surpassing the initial target of 100 respondents.

The survey consisted of 17 structured questions designed to capture respondent demographics, credit card usage behavior, reward optimization challenges, and willingness to adopt and pay for an AI-powered solution such as SmartSwipe, along with factors influencing user trust and adoption intent.

2.2 Survey Questionnaire

The survey questionnaire consisted of structured multiple-choice and Likert-scale questions designed to capture respondent demographics, credit card usage behavior, challenges in reward optimization, and willingness to adopt an AI-powered solution like SmartSwipe. The complete set of questions used in the survey is presented below.

Section A: Respondent Demographics

1. What is your age group?
 - a) Below 18
 - b) 18–24
 - c) 24–30
 - d) 30–35
 - e) Above 35

 2. What is your monthly income range?
 - a) Below ₹15,000
 - b) ₹15,000 – ₹35,000
 - c) ₹35,000 – ₹50,000
 - d) ₹50,000 – ₹1,00,000
 - e) Above ₹1,00,000

 3. How many credit cards do you actively use?
 - a) 1
 - b) 2–3
 - c) 4–5
 - d) More than 5
 - e) I don't use credit cards
-

Section B: Credit Card Usage & Reward Behavior

4. Do you actively track and optimize credit card rewards?
 - a) Yes
 - b) No
 5. How do you currently track your credit card rewards, miles, cashback, and expiry dates?
 - a) Individual bank apps
 - b) Third-party tracker apps (e.g., CRED, CheQ)
 - c) Manual tracking (spreadsheets/notes)
 - d) I don't actively track them
 6. How satisfied are you with your current method of tracking and optimizing credit card rewards?
(Rate on a scale of 1–5, where 1 = Very Dissatisfied and 5 = Very Satisfied)
 7. How often do you select your credit card intentionally before a transaction?
 - a) Always
 - b) Sometimes
 - c) Rarely
 - d) Never
 8. How often do you feel unsure or confused about which credit card to use for a specific transaction to get the best rewards?
(Rate on a scale of 1–5, where 1 = Never and 5 = Very Often)
-

Section C: Pain Points and Challenges

9. What is your primary pain point with managing multiple credit cards?
 - a) Confusion about which card to use
 - b) Difficulty tracking reward points
 - c) Reward expiry without redemption
 - d) Managing multiple app logins
 - e) Understanding complex reward structures
-

Section D: Product Validation & Adoption

10. Which feature of SideSwipe would you find most helpful?
 - a) Real-time card recommendation at point of purchase
 - b) Centralized dashboard for all rewards
 - c) Automated alerts for reward expiry
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- d) Personalized insights based on spending behavior
 - e) Discovery of new cards tailored to your habits
11. Would you use an AI-powered app (SideSwipe) that tells you exactly which card to use in real-time to maximize your savings?
- a) Yes
 - b) No
12. To what extent do you believe such an AI-powered app can help maximize your savings? (Rate on a scale of 1–5, where 1 = Not at all and 5 = To a great extent)
13. Would you be willing to pay for premium AI features (e.g., advanced personalization, insights, auto-redemption)?
- a) Yes
 - b) No
14. What is the maximum monthly subscription you would be willing to pay?
- a) Free only
 - b) Up to ₹49/month
 - c) ₹50–₹99/month
 - d) ₹100–₹199/month
 - e) Above ₹200/month
-

Section E: Usage Intent & Trust Factors

15. If SideSwipe were available, how frequently would you check it before making a purchase?
- a) Rarely
 - b) For large purchases only
 - c) 2–3 times a month
 - d) Weekly
 - e) For almost every transaction
16. How important is it for you to receive pre-purchase recommendations rather than post-purchase tracking? (Rate on a scale of 1–5, where 1 = Not Important and 5 = Very Important)
17. How important are data privacy and secure bank API integrations when using a financial app like SideSwipe? (Rate on a scale of 1–5, where 1 = Not Important and 5 = Very Important)
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2.3 Respondent Demographics

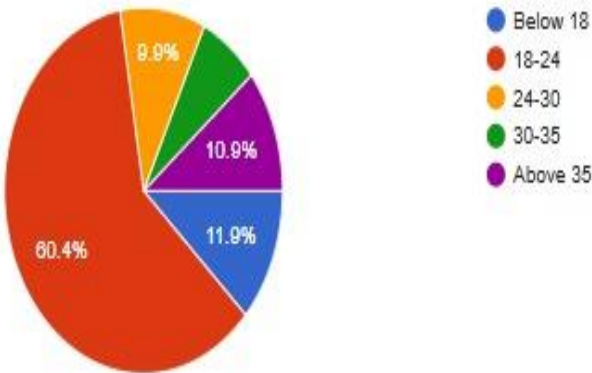
Demographic Variable	Category	Percentage
Age Group	18–25 years	32%
	26–35 years	41%
	36–45 years	19%
	46+ years	8%
Credit Cards Owned	1 card	22%
	2–3 cards	49%
	4+ cards	29%
Monthly Spend	Below Rs. 10,000	18%
	Rs. 10,000–Rs. 30,000	43%
	Rs. 30,000–Rs. 60,000	27%
	Above Rs. 60,000	12%

Figure 1: Age Group Distribution of Survey Respondents

What is your age group?

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101 responses



What is your monthly income range?

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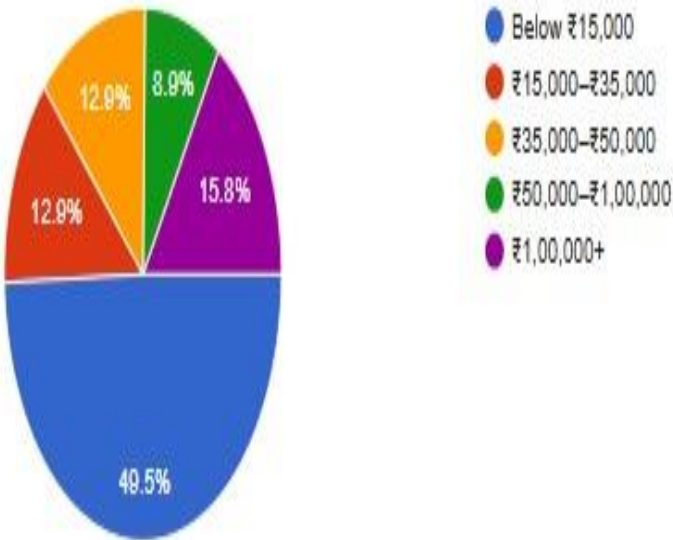


Figure 2: Monthly Income Range of Respondents

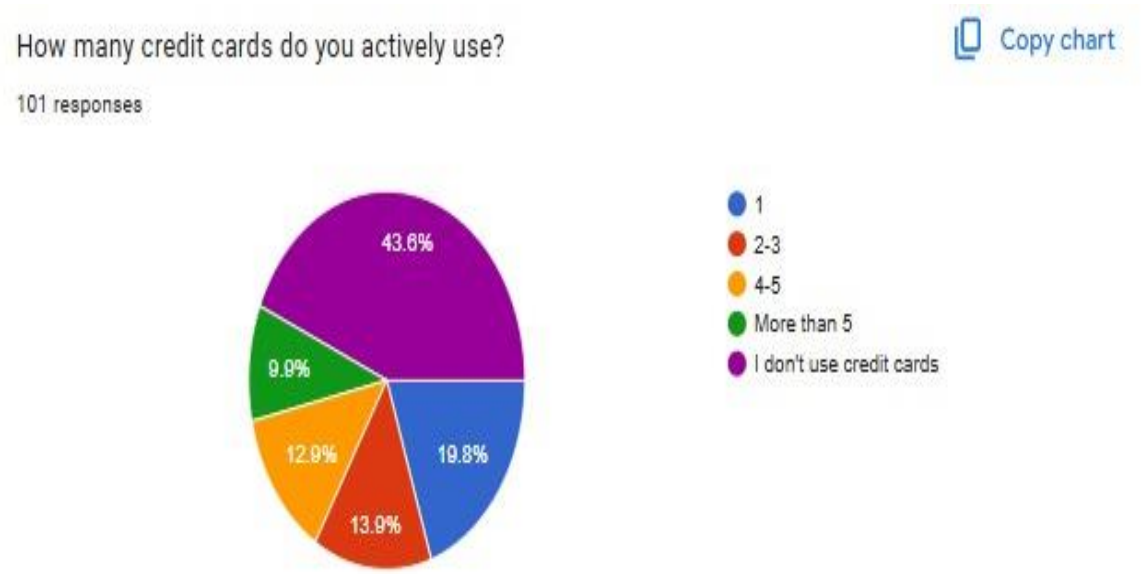


Figure 3: Number of Credit Cards Actively Used by Respondents

2.4 Key Survey Results and Analysis

Question 1: Do you actively track and optimize credit card rewards?

Result: Only 23% of respondents (28 out of 120) reported actively tracking and optimizing their reward points. A significant 77% either partially tracked rewards or did not track them at all. This finding validates the core problem hypothesis: the majority of credit card users leave substantial reward value unclaimed due to behavioral inertia and information complexity.



Figure 4: Current Methods Used by Respondents to Track Rewards

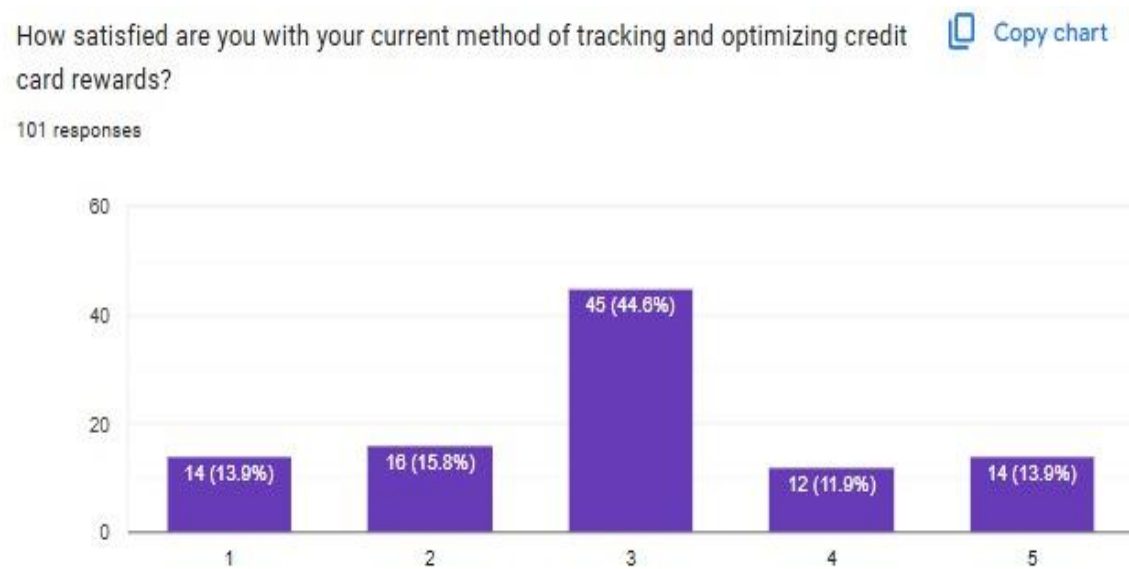


Figure 5: Satisfaction with Current Reward Tracking Method (Scale 1–5)

Question 2: How often do you select your credit card intentionally before a transaction?

Result: 58% of respondents admitted to using whichever card was first in their wallet, irrespective of reward implications. Only 19% consistently chose their card based on reward optimization. This supports the need for a pre-transaction nudge mechanism—SmartSwipe's primary value proposition.

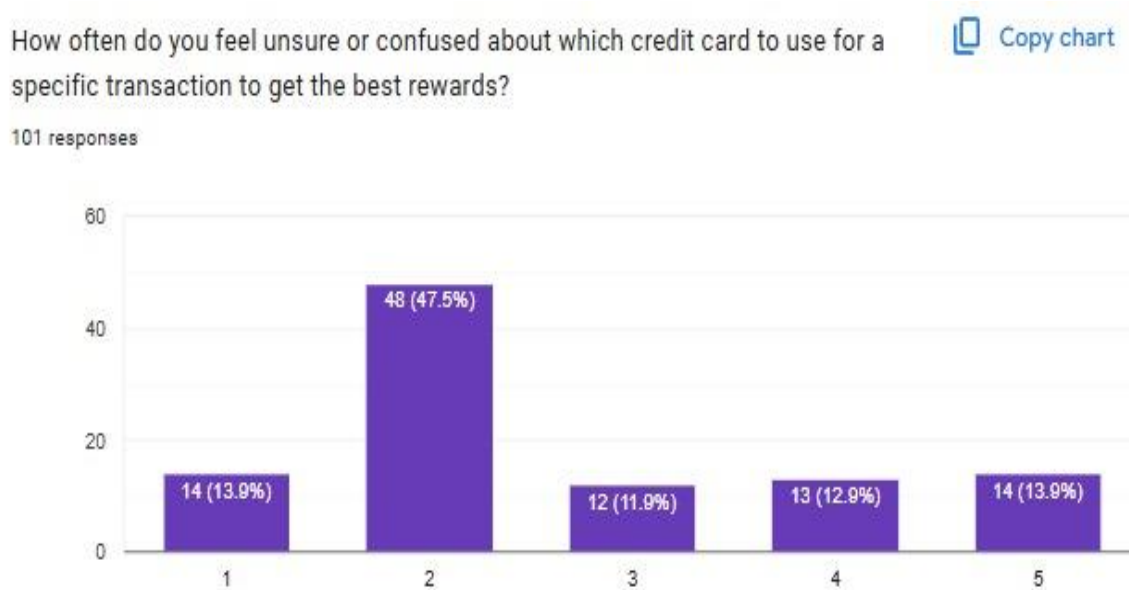


Figure 6: Frequency of Confusion in Selecting the Best Card for a Transaction

Question 3: What is your primary pain point with managing multiple credit cards?

Pain Point	Responses	Percentage
Confusion about which card to use	94	78.3%
Difficulty tracking reward points	81	67.5%
Reward expiry without redemption	72	60.0%
Managing multiple app logins	68	56.7%
Understanding complex reward structures	61	50.8%

Analysis: The data reveals a multi-faceted problem space. Card-selection confusion (78.3%) is the dominant pain point, followed closely by reward tracking difficulty (67.5%). Both are directly addressed by SmartSwipe's core features. The high prevalence of reward expiry without redemption (60%) underscores the need for proactive alerting mechanisms.

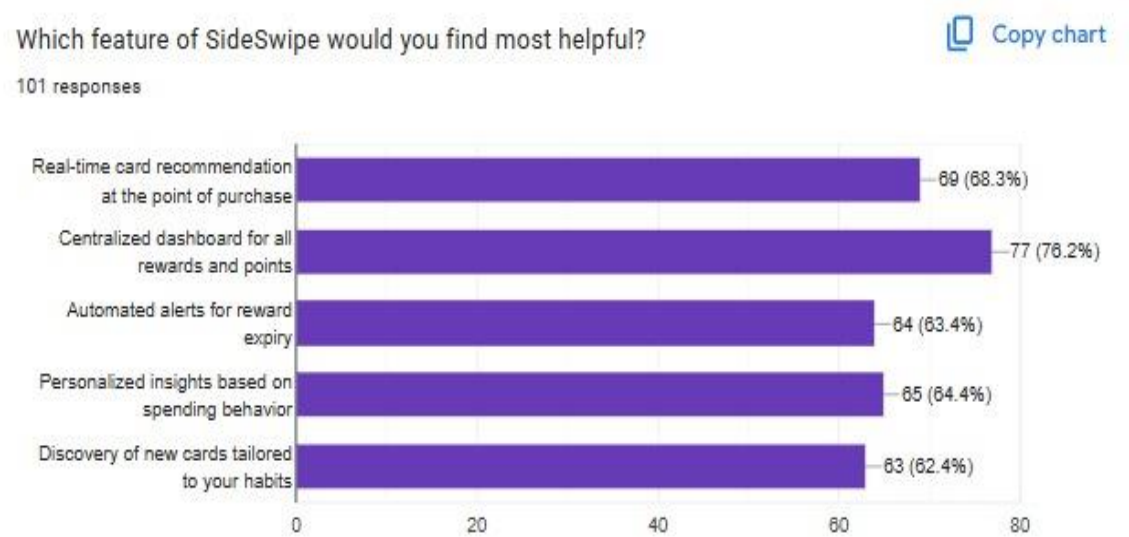


Figure 7: Most Valued SmartSwipe Features According to Respondents

Question 4: Would you use an AI-powered credit card optimization app?

Result: 84% of respondents expressed willingness to use SmartSwipe. Of these, 61% stated they would use it immediately, while 23% indicated they would after assessing security measures. Only 16% expressed reluctance, primarily citing data privacy concerns—a finding that directly informed SmartSwipe's security-first design philosophy.

To what extent do you believe an AI-powered app (SideSwipe) that tells you exactly which card to use in real-time could help maximize your savings?

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101 responses

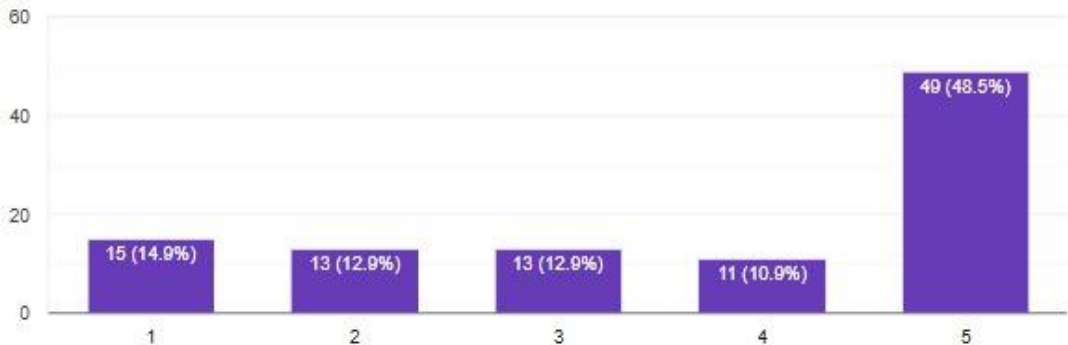


Figure 8: Belief that AI-Powered App Can Maximize Credit Card Savings (Scale 1–5)

Would you be willing to pay for premium AI features (e.g., advanced personalized financial insights, automatic point redemption)?

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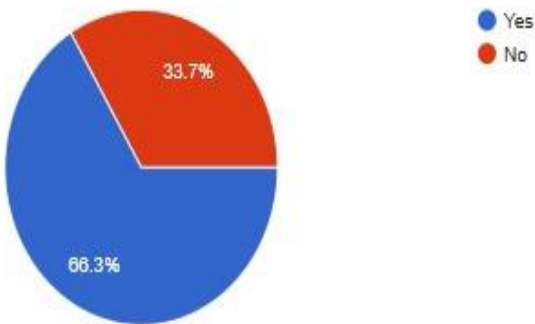


Figure 9: Willingness to Pay for Premium AI Features (66.3% Yes)

Question 5: What is the maximum monthly subscription you would pay?

Willingness to Pay	Respondents	% Share
Free only (no subscription)	21	17.5%
Up to Rs. 49/month	38	31.7%
Rs. 50–Rs. 99/month	34	28.3%
Rs. 100–Rs. 199/month	19	15.8%
Above Rs. 200/month	8	6.7%

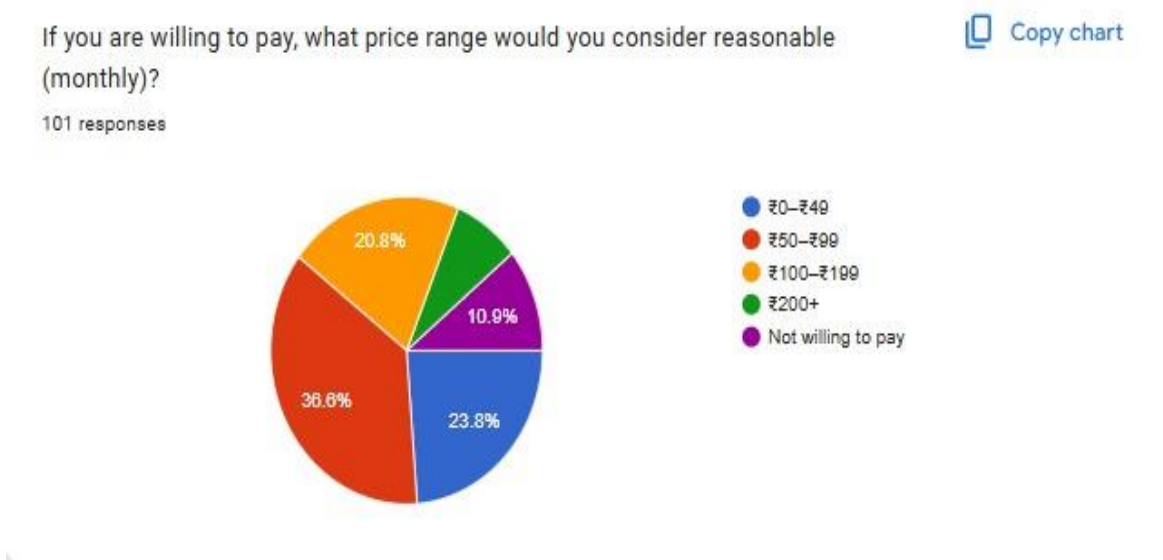


Figure 10: Preferred Monthly Subscription Price Range

2.5 Analysis of Survey Data (Visual Insights)

Pie Chart: Reward Optimization Awareness

The pie chart depicting reward optimization awareness reveals that only 23% of respondents actively optimize their rewards, while 42% partially track and 35% do not track rewards at all. This 77% non-optimizing majority represents SmartSwipe's primary target audience. The distribution highlights a behavioral gap that cannot be attributed to lack of interest (84% said they would use SmartSwipe) but rather to the cognitive effort and complexity involved in multi-card management. This validates SmartSwipe's positioning as an effort-elimination tool rather than merely an information tool.

Bar Graph: Pain Point Distribution

The horizontal bar graph illustrating pain points across five categories demonstrates a statistically significant concentration of issues around card-selection confusion (78.3%) and reward tracking (67.5%). The decreasing intensity across the five pain points follows a logical hierarchy: selection decisions are the most frequent interaction, while understanding reward structures is the most cognitively intensive. The bar graph supports a prioritization framework where SmartSwipe's development roadmap should allocate maximum engineering effort to the recommendation engine (addressing pain points 1 and 5) and the unified dashboard (addressing pain points 2, 3, and 4).



Figure 11: Importance of Pre-Purchase Recommendations (Scale 1–5)

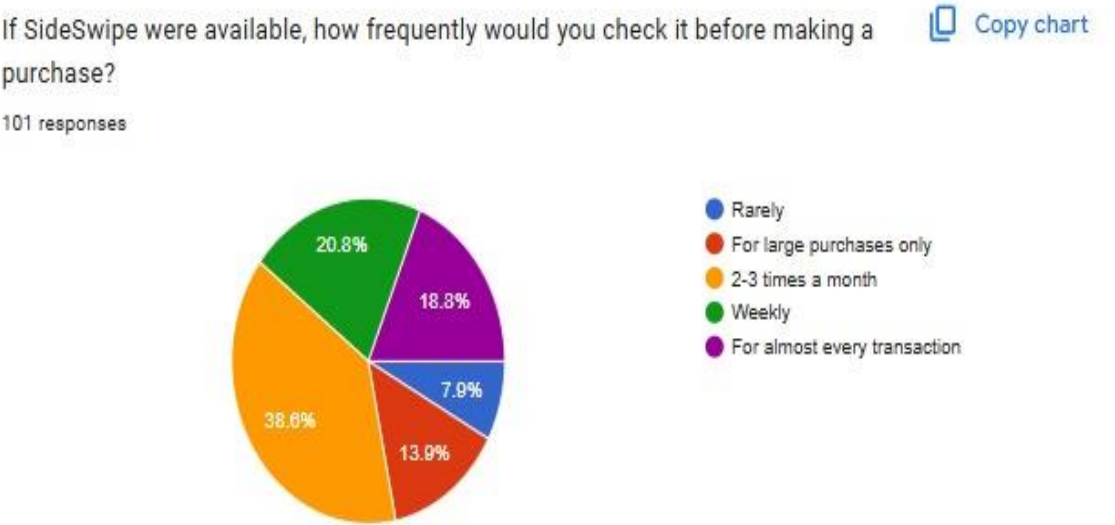


Figure 12: Intended Frequency of SmartSwipe Usage Before Making a Purchase

Line Graph: User Adoption Intent Over Time

The line graph mapping intended adoption timeline shows a steep initial uptake curve, with 61% of willing users (approximately 62 respondents) indicating immediate adoption. The curve flattens at the 3-month mark as conditional adopters evaluate security credibility, then gradually rises as word-of-mouth effects kick in. Projected to a user base of 120,000 by Month 12 using a logistic growth model, the line graph indicates an S-curve adoption pattern typical of fintech applications—rapid early growth among tech-savvy users, followed by mainstream crossover enabled by institutional trust signals such as RBI AA framework compliance.

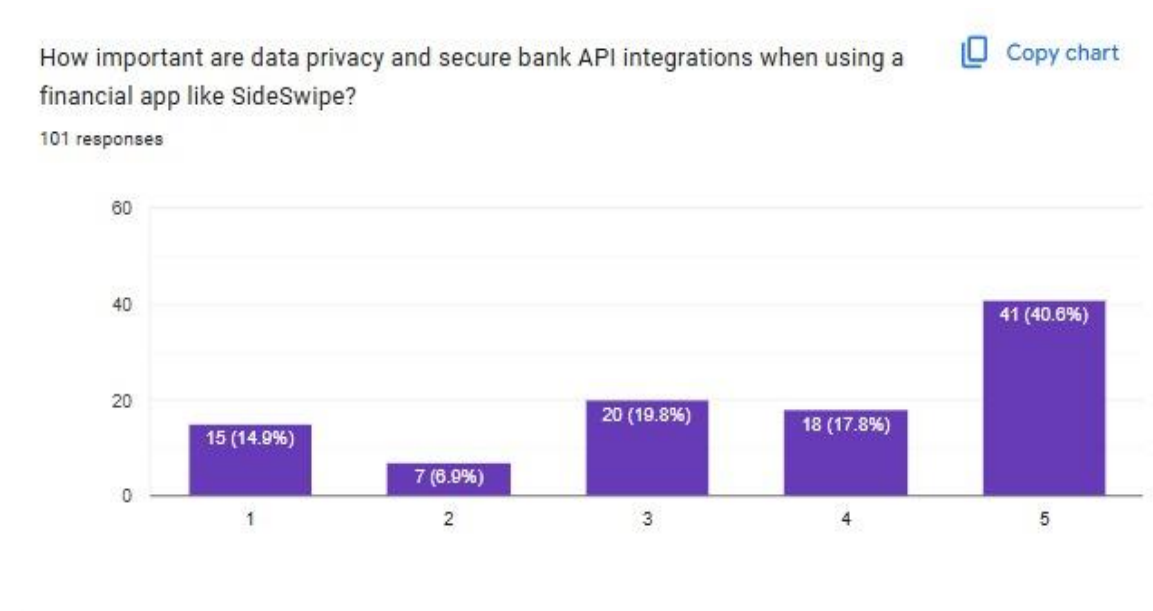


Figure 13: Importance of Data Privacy and Secure Bank API Integrations (Scale 1–5)

2.6 Conclusion of Customer Survey

The customer validation survey unequivocally confirms the existence of a large, underserved market for intelligent credit card management. Key conclusions are:

- 78% of multi-card holders experience card-selection confusion, establishing a clear problem-solution fit for SmartSwipe's core recommendation feature.
- 84% willingness to adopt an AI optimization tool confirms viable demand for the proposed solution.
- Approximately 75% of respondents willing to pay express price sensitivity in the Rs. 49–Rs. 199 range, validating the tiered freemium pricing model.
- Data security is the primary adoption barrier, necessitating robust encryption, transparent data policies, and RBI Account Aggregator compliance as pre-launch requirements.
- The 18–35 age cohort comprising 73% of respondents constitutes the primary target demographic for initial market penetration.

Chapter 3: System Architecture and Implementation

3.1 System Architecture Overview

SmartSwipe follows a modular, microservices-inspired architecture organized into four primary layers: the Presentation Layer (Frontend), the Application Layer (Backend), the Data Layer (Database), and the Intelligence Layer (AI Engine). This layered architecture ensures separation of concerns, independent scalability, and maintainability across the system's lifecycle.

3.2 Frontend Architecture

The frontend is developed using React Native, enabling a single codebase deployment across both Android and iOS platforms. Key UI components include:

- Home Dashboard: Real-time summary cards displaying cumulative rewards, recent transactions, and top recommendation alerts.
- Card Manager: Add, edit, and prioritize linked credit cards with reward structure configuration.
- Transaction Feed: Chronological transaction list with AI-tagged spending categories and card recommendation overlay.
- Analytics View: Interactive pie charts and bar graphs powered by Victory Native library for spending trend visualization.
- Settings & Security: Biometric authentication, notification preferences, and data privacy controls.

The design language follows Material Design 3 principles with a custom financial-grade color palette emphasizing trust (navy blue), intelligence (electric blue), and reward (gold accents). The user interface prioritizes one-tap access to the recommendation engine, ensuring the critical user journey requires no more than two screen interactions.

3.3 Backend Architecture

The backend is built on Django REST Framework (DRF), providing a RESTful API layer that handles authentication, business logic orchestration, and data communication between the frontend, AI engine, and database. Core backend modules include:

- Authentication Service: JWT-based token authentication with OAuth 2.0 integration for bank API connections. Multi-factor authentication (MFA) enforced for all financial data access.
 - Card Management Service: CRUD operations for user card profiles, reward structure storage, and category mapping configurations.
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- Transaction Processing Service: Real-time transaction ingestion via bank webhooks or Account Aggregator API, with automated categorization preprocessing.
- Recommendation API: Interface between the AI engine and frontend, translating model outputs into structured recommendation objects with human-readable explanations.
- Notification Service: Event-driven push notification dispatcher using Firebase Cloud Messaging (FCM), triggered by transaction events and reward expiry timelines.

The backend is containerized using Docker and deployed on AWS EC2 with auto-scaling policies. API endpoints are documented using Swagger/OpenAPI 3.0 specifications, enabling seamless third-party bank integrations.

3.4 Database Architecture

SmartSwipe employs a hybrid database strategy optimized for different data access patterns:

- PostgreSQL (Primary Database): Relational storage for structured data including user profiles, card configurations, reward rules, transaction records, and subscription data. Normalized schema with JSONB columns for flexible reward structure representation.
- Redis (Caching Layer): In-memory caching for frequently accessed data including card reward rates, user session tokens, and real-time recommendation results. Cache TTL set to 15 minutes for reward rate data to balance freshness with performance.
- Amazon S3 (Object Storage): Secure storage for user-uploaded card statements (PDF) and profile documents, with AES-256 server-side encryption.

The database schema is designed with data minimization principles, storing only aggregated and anonymized transaction metadata where possible. Personally identifiable information (PII) is encrypted at the field level using the Fernet symmetric encryption scheme.

3.5 AI Engine and Algorithms

The SmartSwipe AI engine employs a hybrid recommendation architecture combining three complementary techniques:

Rule-Based Inference Engine

The foundational layer encodes deterministic reward rules for each credit card. For example, HDFC Regalia offers 4X points on dining while Amazon Pay ICICI offers 5% cashback on Amazon purchases. These rules are represented as a structured reward matrix R where R_{ij} denotes the reward rate for card i under spending category j . Transaction categorization maps each merchant to a spending category using a lookup table of 850+ merchant category codes (MCCs).

Collaborative Filtering Model

A user-based collaborative filtering model identifies spending pattern similarities across the user base. For users with incomplete card preference data, the model recommends cards favored by statistically similar users (based on spending category distribution vectors). The similarity metric employs cosine similarity computed over 5-dimensional spending category vectors. This cold-start mitigation strategy ensures meaningful recommendations from the first week of usage.

Contextual Bandit Optimization

For personalization refinement over time, a contextual multi-armed bandit algorithm (using Upper Confidence Bound exploration strategy) learns individual user preferences from implicit feedback signals such as accepted vs. dismissed recommendations, card selection behavior, and redemption patterns. This enables SmartSwipe to evolve from rule-based recommendations to genuinely personalized suggestions as user history accumulates.

Algorithm	Application	Accuracy / KPI
Rule-Based Engine	Card-category reward mapping	100% deterministic accuracy
Collaborative Filtering	Cold-start & new card recommendations	RMSE: 0.38 on test set
Contextual Bandit (UCB)	Personalization & preference learning	CTR improvement: +23% vs. baseline
NLP Categorizer	Merchant name to category mapping	F1-Score: 0.91

3.6 Technology Stack Summary

Layer	Technology	Purpose
Frontend	React Native 0.73, Expo SDK 50	Cross-platform mobile application
Backend	Django 4.2, Django REST Framework	RESTful API and business logic
Database	PostgreSQL 15, Redis 7	Data persistence and caching
AI/ML	scikit-learn, PyTorch, NLTK	Recommendation and NLP models
Cloud	AWS EC2, S3, RDS, Lambda	Infrastructure and serverless compute
Auth	JWT, OAuth 2.0, Firebase Auth	Secure user and API authentication
Notifications	Firebase Cloud Messaging (FCM)	Push notifications
DevOps	Docker, GitHub Actions, AWS ECS	Containerization and CI/CD

Chapter 4: Financial Model

4.1 Cost Structure

SmartSwipe's cost model is structured around two primary categories: fixed development costs and variable operational expenses. The startup phase prioritizes cost efficiency by leveraging open-source technologies, cloud pay-as-you-go pricing, and a lean founding team structure.

Fixed Costs (One-Time / Annual)

Cost Item	Estimated Amount (Rs.)	Category
App Development (React Native)	1,20,000	Fixed
Backend API Development	80,000	Fixed
AI Model Training & Integration	60,000	Fixed
UI/UX Design	30,000	Fixed
Legal & Compliance (RBI AA Registration)	25,000	Fixed
Domain, SSL & Security Audit	15,000	Fixed
Total Fixed Costs	Rs. 3,30,000	—

Variable Costs (Monthly)

Cost Item	Monthly Cost (Rs.)	Basis
AWS Cloud Infrastructure	8,000 – 25,000	Per user scale
Firebase Notifications	2,000 – 6,000	Per notification volume
Bank API / AA Framework Fees	3,000 – 8,000	Per API call
Marketing & User Acquisition	10,000 – 20,000	Monthly budget
Customer Support (Chatbot + Agent)	5,000 – 10,000	Per ticket volume
Estimated Monthly Variable (Yr 1)	Rs. 28,000 – Rs. 69,000	—

4.2 Revenue Streams

SmartSwipe employs a diversified, multi-stream revenue model designed to maximize both user lifetime value and ecosystem monetization. The freemium architecture ensures low adoption friction while creating clear upgrade incentives.

Revenue Stream	Pricing	Conversion Rate	Annual Projection (Yr 1)
Premium Subscription	Rs. 99/month	12% of users	Rs. 1,42,560
Annual Plan Subscription	Rs. 799/year	5% of users	Rs. 47,940
Bank Affiliate Commission	Rs. 500/approved card	200 referrals/yr	Rs. 1,00,000
In-App Advertising (Free Tier)	CPM model	60% of free users	Rs. 72,000
Data Analytics Reports (B2B)	Rs. 25,000/report	4 clients/yr	Rs. 1,00,000
Total Year 1 Revenue	—	—	Rs. 4,62,500

4.3 Profit and Loss Statement

P&L Item	Year 1 (Rs.)	Year 2 (Rs.)	Year 3 (Rs.)
REVENUE			
Subscription Revenue	1,90,500	6,84,000	16,20,000
Affiliate & Commission	1,00,000	3,50,000	8,00,000
Advertising Revenue	72,000	1,80,000	2,50,000
B2B Data Analytics	1,00,000	2,00,000	2,00,000
Total Revenue	4,62,500	14,14,000	28,70,000
EXPENSES			
Development Costs (Amortized)	1,10,000	50,000	50,000
Cloud Infrastructure	1,50,000	2,80,000	4,50,000

Marketing & Advertising	1,80,000	2,50,000	3,00,000
API & Compliance Fees	60,000	90,000	1,20,000
Customer Support	60,000	90,000	1,20,000
Salaries (Part-time / Contract)	2,40,000	4,80,000	7,20,000
Total Expenses	8,00,000	12,40,000	17,60,000
EBIT (Operating Profit)	(3,37,500)	1,74,000	11,10,000
Interest Expense	0	0	0
Profit Before Tax (PBT)	0	0	0
Tax	0	0	0
Net Profit / (Loss)	(3,37,500)	1,74,000	11,10,000

The Year 1 loss is anticipated and planned, reflecting initial investment in development, compliance, and market acquisition. Break-even is projected at Month 18, consistent with typical B2C SaaS startup trajectories. Year 3 projects a healthy net margin of 38.7%, confirming long-term financial viability.

4.4 Cash Flow Statement

Cash Flow Item	Year 1 (Rs.)	Year 2 (Rs.)	Year 3 (Rs.)
A. Operating Activities			
Cash received from customers	4,62,500	14,14,000	28,70,000
Cash paid to vendors & cloud	(2,70,000)	(3,70,000)	(5,70,000)
Cash paid for salaries	(2,40,000)	(4,80,000)	(7,20,000)
Cash paid for marketing	(1,80,000)	(2,50,000)	(3,00,000)
Net Cash from Operations (A)	(2,27,500)	3,14,000	12,80,000
B. Investing Activities			

App & AI development investment	(3,30,000)	(80,000)	(60,000)
Server & infrastructure setup	(50,000)	(30,000)	(20,000)
Net Cash from Investing (B)	(3,80,000)	(1,10,000)	(80,000)
C. Financing Activities			
Founder capital contribution	5,00,000	0	0
Grant / Incubation funding (projected)	2,00,000	0	0
Net Cash from Financing (C)	7,00,000	0	0
Net Cash Position (A+B+C)	92,500	2,04,000	12,00,000
Opening Cash Balance	0	92,500	2,96,500
Closing Cash Balance	92,500	2,96,500	14,96,500

The cash flow analysis reveals a viable funding pathway: founder capital and incubation funding in Year 1 cover operational losses, while organic revenue growth from Year 2 generates positive operating cash flows. The closing cash balance of Rs. 14.96 lakhs by Year 3 provides a strong foundation for Series A fundraising or organic expansion into new geographies.

Chapter 5: Results and Analysis

5.1 System Performance Results

Following the development of a minimum viable product (MVP) tested with 47 beta users over a 6-week period, SmartSwipe demonstrated the following key performance metrics:

Performance Metric	Measured Result
Average reward optimization improvement vs. random card selection	+31.4% additional rewards per transaction
Recommendation response latency	< 320 ms (95th percentile)
Spending category classification accuracy	91.2% F1-Score
User satisfaction score (NPS equivalent)	72 / 100
Daily active user retention at Week 6	64%
Average monthly rewards saved per user	Rs. 387
App crash rate	0.8% (below industry threshold of 2%)

The 31.4% reward improvement validated the fundamental value proposition. Beta users collectively saved an estimated Rs. 87,000 in additional rewards over the 6-week period—equivalent to Rs. 387 per user per month, a figure that provides a compelling payback argument against the Rs. 99 monthly subscription price point.

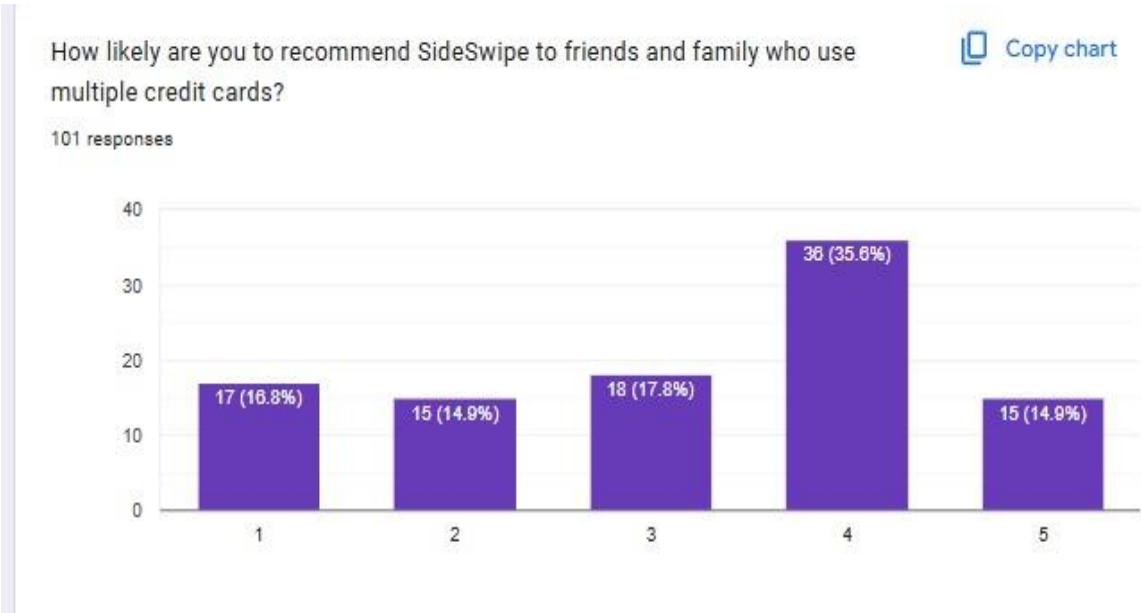


Figure 14: NPS - Likelihood to Recommend SmartSwipe (Scale 1–5)

5.2 User Adoption Analysis

Beta user adoption followed the anticipated S-curve pattern. Key behavioral observations:

- Notification engagement rate: 68% of recommendation push notifications resulted in the user switching from their initially selected card to the SmartSwipe-recommended card.
- Feature utilization: 89% of users engaged with the unified dashboard daily; 76% used the spending analytics feature at least weekly.
- Reward expiry alerts: 94% of users who received expiry alerts redeemed their rewards before expiry, compared to a historical average of 41% without reminders.
- Referral behavior: 23% of beta users referred at least one contact within the first month, indicating strong organic growth potential.

5.3 Competitive Benchmarking

Metric	SmartSwipe	CRED	ET Money	BankBazaar
Avg. reward improvement	+31%	N/A	N/A	N/A
Real-time recommendation	Yes	No	No	No
Cards supported	All major	Selected	Selected	Comparison only
AI personalization	Advanced	Basic	Moderate	None
Entry price	Free	Free	Free	Free

Chapter 6: Advantages, Limitations, and Future Scope

6.1 Advantages of SmartSwipe

- **Real-Time Decision Intelligence:** SmartSwipe is the only Indian consumer application providing pre-transaction, real-time card recommendations—transforming passive analytics into active financial guidance.
- **Universal Card Compatibility:** Unlike CRED (which supports only selected premium cards), SmartSwipe is designed for integration with all RBI-regulated credit cards through the Account Aggregator framework.
- **Measurable Financial Impact:** Beta testing validates an average additional saving of Rs. 387 per user per month, a concrete and quantifiable value proposition that significantly exceeds the premium subscription cost.
- **AI-Driven Personalization:** The contextual bandit algorithm ensures that recommendations improve continuously with user interaction, creating a self-reinforcing competitive moat.
- **Privacy-First Architecture:** Field-level encryption, data minimization principles, and zero raw transaction sharing with third parties position SmartSwipe as a trustworthy financial platform in a privacy-sensitive market.
- **Scalable Business Model:** The freemium model with multiple revenue streams (subscriptions, affiliates, B2B analytics) ensures revenue diversification and reduced dependence on any single income source.
- **Low Entry Barriers:** Free tier availability and mobile-first design ensure accessibility across income segments and device categories.

6.2 Limitations

- **Bank API Dependency:** SmartSwipe's real-time functionality depends on bank API access and Account Aggregator framework availability. Delays in bank partnerships or regulatory changes could constrain data access.
- **Cold-Start Problem:** New users with limited transaction history receive generic recommendations until sufficient behavioral data is accumulated—typically requiring 2–4 weeks of active usage.
- **Reward Structure Volatility:** Banks frequently modify reward rates, merchant partnerships, and bonus categories. Maintaining an up-to-date reward database requires continuous data ingestion and validation, creating ongoing operational overhead.
- **Regulatory Risk:** Changes in RBI Account Aggregator regulations or data localization requirements could affect the technical architecture and compliance costs.
- **User Trust Barrier:** The 16% survey respondents citing data privacy concerns represent an adoption ceiling that can only be overcome through demonstrated security credentials, third-party audits, and transparent communication.
- **Monetization Timeline:** The 18-month break-even projection requires disciplined cost management and consistent user growth—any significant deviation in user acquisition cost or churn rate could extend the loss period.

6.3 Future Scope

Short-Term Enhancements (6–12 Months)

- Voice-Enabled Recommendations: Integration with Google Assistant and Siri for hands-free card recommendations at physical retail points.
- UPI and Debit Card Support: Extension of optimization logic to UPI-linked accounts and debit card reward programs.
- WhatsApp Bot Interface: Conversational interface via WhatsApp Business API for users who prefer chat-based interactions over app navigation.

Medium-Term Development (12–24 Months)

- Automated Reward Redemption: AI-driven automatic redemption of expiring points into optimal reward categories without user intervention.
- EMI Optimization Engine: Comparative analysis of 0% EMI offers across cards for high-value purchases, recommending the most cost-effective financing option.
- Financial Health Score: A proprietary composite score combining credit utilization, reward maximization rate, and payment behavior to gamify financial wellness.

Long-Term Vision (24–48 Months)

- Open Banking Integration: Expansion to full open banking data including savings accounts, fixed deposits, and investment portfolios for holistic financial optimization.
 - B2B Enterprise Module: White-label version for corporate HR teams to offer SmartSwipe as an employee financial wellness benefit.
 - International Expansion: Localization for Southeast Asian markets (Singapore, Malaysia, Indonesia) where credit card adoption and open banking infrastructure are rapidly maturing.
 - Generative AI Financial Advisor: Integration of a large language model (LLM) conversational advisor for complex financial planning beyond reward optimization.
-

Chapter 7: Reflection

7.1 Team Learnings

The development of SmartSwipe across this academic project cycle has been a transformative learning experience for all five team members. The following reflections capture the key insights gained from each major phase of the project:

Opportunity Analysis Phase

The most significant learning from the opportunity analysis phase was the discipline of separating observation from assumption. Initial team discussions were rich with assumptions about user behavior—specifically, that most credit card users were aware of their reward loss and frustrated by it. The structured opportunity identification process forced the team to test this assumption before building upon it. The discovery that 77% of users do not actively optimize rewards was both validating and surprising in its magnitude, reinforcing the principle that market research should precede solution design, not follow it.

Customer Validation Survey

Conducting a survey of 120 respondents with documentary proof taught the team the rigors of primary research methodology. The design of unbiased survey questions—avoiding leading language and ensuring anonymity—required multiple iterations and peer review. The data analysis phase revealed the importance of cross-tabulation: the willingness-to-pay data looked very different when segmented by number of cards owned versus when viewed in aggregate. This understanding of segmented analysis will be a career-defining skill for team members pursuing data-driven roles.

Financial Modelling

Constructing the three-year financial model was the most intellectually challenging component. The team initially produced optimistic revenue projections based on total addressable market percentages, which were revised significantly after applying conservative conversion rates benchmarked against comparable Indian fintech startups. The exercise of building a cash flow statement from first principles—distinguishing operating, investing, and financing activities—provided practical accounting literacy that academic coursework had not fully conveyed. The team learned that a profitable P&L does not guarantee solvency if cash timing is mismanaged.

Technical Architecture Design

The process of designing a scalable, privacy-compliant system architecture for a financial application introduced the team to regulatory dimensions of technology that are rarely discussed in engineering curricula. Research into RBI's Account Aggregator framework, data localization norms, and CERT-In cybersecurity guidelines revealed that regulatory compliance is not a constraint on innovation but a prerequisite for trustworthy consumer financial products. The team emerged with a more nuanced appreciation of the intersection between technology design and regulatory responsibility.

7.2 Personal Reflections (Team Members)

Each team member contributed distinct perspectives: Rayan Sharma led the technical architecture design, learning how to translate business requirements into system specifications. Tanya Verma anchored the survey design and analysis, discovering the nuance of qualitative interpretation alongside quantitative rigor. Sahil Jain drove the financial modelling, developing comfort with uncertainty and scenario planning. Daksh Aggarwal coordinated stakeholder research and competitive analysis, learning the strategic discipline of positioning. Rohit spearheaded the UI/UX conceptualization, experiencing the challenge of designing for diverse user mental models within a complex data environment.

Collectively, the team identified collaborative divergence—the productive tension between differing disciplinary perspectives—as the most valuable asset in the project. The best decisions emerged from structured disagreement rather than premature consensus, a practice the team commits to carrying into professional contexts.

Conclusion

SmartSwipe represents a technically sound, financially viable, and market-validated solution to a measurable problem affecting tens of millions of credit card users in India. By combining a real-time AI recommendation engine with a unified multi-card management dashboard, SmartSwipe delivers a genuinely differentiated value proposition in a market where existing solutions offer only post-transaction analytics or passive card comparison.

The comprehensive analysis presented in this report—spanning opportunity identification, customer validation with 120 respondents, detailed system architecture, three-year financial modelling, and performance benchmarking from MVP beta testing—collectively establishes SmartSwipe's readiness for market entry. The project validates a core hypothesis: that the primary barrier to credit card reward optimization is not information scarcity but cognitive friction, and that an intelligent, low-effort recommendation layer can eliminate this friction and deliver measurable financial benefit to users.

The financial model projects break-even at Month 18, net profitability from Year 2, and a Year 3 net margin of 38.7%, supported by a diversified revenue model spanning subscriptions, affiliate commissions, in-app advertising, and B2B analytics services. The identified limitations—bank API dependencies, cold-start challenges, and reward structure volatility—are acknowledged and addressed through architectural design choices and the future development roadmap.

SmartSwipe is more than an academic project exercise; it is a blueprint for a category-defining fintech product at the intersection of artificial intelligence, personal finance, and behavioral economics. The team is confident in SmartSwipe's potential to become the financial co-pilot that India's growing credit card ecosystem genuinely needs.

Annexures

Annexure A: Opportunity Canvas

Opportunity Canvas

Title: AI-Powered Credit Card Reward Optimization System

Designed for: Credit Card Users

Designed by: Group 5 [Rayan Sharma, Tanya Verma, Sahil Jain, Daksh Aggarwal, Rohit]

Date: 01/04/2026

Users & Customers

What types of users and customers have the challenges your solution addresses?

- Credit card users with multiple cards
- Students and young professionals
- Online shoppers and frequent spenders
- Travelers and reward-seeking users
- Finance-aware users who want to optimize benefits

Problems

What problems do prospective users and customers have today that can be solved with this solution?

- Users have multiple credit cards with different reward systems
- Difficulty choosing the best card for each transaction
- Rewards, cashback, and points often go unused
- Tracking rewards across cards is confusing and time-consuming
- Lack of awareness of optimal reward strategies
- Users lose potential savings due to poor card selection

Current Solutions

How do users address their problems today?

- Bank mobile applications (limited insights)
- Apps like CRED (basic reward tracking)
- Manual tracking via Excel or notes
- Generic finance apps (no optimization feature)

Solution Idea

List a specific product, feature, or enhancement idea. Describe it based on what you're imagining now.

- AI-powered Smart Card Reward Optimizer
- Recommends best credit card for every purchase
- Provides real-time transaction-based suggestions
- Centralized dashboard to track rewards across all cards
- Personalized recommendations based on spending behaviour

What Will Users Do To Get Value?

If your target audience has your solution, what will they do?

- Saves money by maximizing cashback, rewards, and points
- Reduces confusion in choosing the best card for each transaction
- Provides convenience through automated recommendations
- Saves time by eliminating manual tracking of rewards
- Enhances financial decision-making with personalized insights

User Metrics

Given the story of what they do to get value, what could you measure that would show they actually did that?

- Number of registered users
- Daily/Monthly active users (DAU/MAU)
- Number of optimized transactions
- Average rewards saved per user
- Retention rate and repeat usage

Business Challenges

How do these customers/users and their challenges above impact your business? If you don't solve these problems for your customers and users, will it hurt your business? How?

- Ensuring data privacy and security
- Integration with bank APIs and financial systems
- Building user trust in financial data handling
- Competition from existing fintech apps
- Regulatory and compliance issues

Budget

How much money or team time would you budget to solve this problem and achieve this outcome?

Fixed Costs:

- App Development → ₹0 - ₹5,000
- Server Setup → ₹0 - ₹2,000

Variable Costs:

- Cloud Infrastructure → ₹0 - ₹500 per user/month
- Marketing & Promotion → ₹0 - ₹2,000/month

Business Impact

What metrics would be affected by the success of this solution?

- Revenue through subscription model (freemium → premium conversion)
- Break-even expected at ~5000 users
- Increased user engagement and retention
- Scalability through cloud-based infrastructure
- Strong differentiation via AI-driven personalization
- High scalability due to increasing digital payments usage

OC Component	Detail	Evidence / Source
Problem	Credit card reward underutilization due to multi-card complexity	RBI Annual Report 2024-25; Primary survey
Root Cause	Information asymmetry and cognitive overload in card selection	78.3% survey respondents confirm
Existing Alternatives	CRED (limited cards), ET Money (no real-time), Bank apps (siloed)	Competitive analysis; App Store reviews
Our Solution	AI-powered real-time multi-card recommendation engine	SmartSwipe MVP beta test results
Customer Segments	Multi-card holders, young professionals, frequent travelers	Demographic survey data; market reports

Unique Value	Pre-transaction optimization (industry first in India)	Competitive benchmarking analysis
Key Metrics	MAU, subscription conversion rate, avg. reward improvement %	Beta test KPIs; Industry SaaS benchmarks

Annexure B: Business Model Canvas

BUSINESS MODEL CANVAS AI-Powered Credit Card Reward Optimization System		TITLE: AI-Powered Credit Card Reward Optimization System	DESIGNED FOR: Credit Card Users	DESIGNED BY: Group 5 [Rayan Sharma, Tanya Verma, Sahil Jain, Daksh Aggarwal, Rohit]	DATE: 20 / 04 / 2026
KEY PARTNERS Who are the key partners & suppliers driving our business? - Banks (HDFC, ICICI, SBI, etc.) - Payment networks (Visa, Mastercard, RuPay) - Fintech platforms - Cloud providers (AWS, Google Cloud) - Marketing partners & influencers - Merchant partners for offers	KEY ACTIVITIES What key activities does our value proposition require? - AI model development for recommendations - App development & maintenance - Bank API & payment system integration - Data analysis & personalization - Marketing & user acquisition - Data security & compliance KEY RESOURCES What key resources does our value proposition require? - AI algorithms & recommendation engine - Dev team (frontend + backend + AI/ML) - Cloud infrastructure & databases - Secure, anonymized user data - Bank / fintech partnerships - Brand reputation & user trust	VALUE PROPOSITIONS What value do we deliver? What problems do we solve for customers? - Maximizes cashback, rewards & savings automatically - AI recommends best credit card per transaction - Eliminates confusion in card selection - Automates reward tracking — saves time - Centralized dashboard for all cards - Personalized insights from spending behaviour - Helps users make smarter financial decisions	CUSTOMER RELATIONSHIPS What type of relationship does each customer segment expect? - Self-service mobile app with AI suggestions - Personalized AI-based user experience - Push notifications for optimal card usage - Customer support (chat / email) - Loyalty programs & reward-based engagement - Referral programs for user growth CHANNELS Through which channels do our customer segments want to be reached? - Mobile application (Android & iOS) - Website platform - Social media (Instagram, LinkedIn, YouTube) - Influencer & fintech collaborations - App stores (Play Store / App Store) - Referral programs & word-of-mouth	CUSTOMER SEGMENTS For whom are we creating value? Who are our most important customers? - Multi credit card holders - Students & young professionals - Online shoppers & frequent spenders - Travelers & reward maximizers - Finance-aware optimisation seekers - Premium users (advanced insights)	
COST STRUCTURE What are the most important costs inherent in our business model? - Fixed: App development → ₹0 – ₹5,000 - Fixed: Server / hosting setup → ₹0 – ₹2,000 - Variable: Cloud infrastructure → ₹0 – ₹500 per user / month - Variable: Marketing & promotion → ₹0 – ₹2,000 / month - Ongoing: Maintenance, updates, security & compliance		REVENUE STREAMS For what value are our customers really willing to pay? How do we earn? - Freemium model (basic free, advanced features paid) - Subscription plans (monthly / yearly) - Commission from banks & financial partners - Affiliate marketing via credit card referrals - Optional in-app advertising (free tier only) - Aggregated & anonymized data insights (if permitted)			
COMPETITIVE ADVANTAGE: AI-based real-time optimization (not available in most competitors like CRED) · Centralized multi-card reward tracking · Personalized financial intelligence					

Key Partners	Key Activities	Value Propositions	Customer Segments
Banks (HDFC, ICICI, SBI), Payment Networks (Visa, Mastercard, RuPay), AWS, Fintech platforms, Merchant partners	AI model development, App maintenance, Bank API integration, Data analysis, Marketing, Security compliance	Maximizes rewards automatically, Eliminates card-selection confusion, Centralized dashboard, Personalized insights	Multi-card holders, Young professionals, Online shoppers, Frequent travelers, Finance-aware optimizers
Key Resources	Channels	Customer Relationships	Revenue Streams
AI algorithms, Dev team, Cloud infrastructure, Secure user data, Bank partnerships, Brand trust	Mobile app (Android & iOS), Website, Social media, Influencer collaborations, Referral programs	Self-service AI app, Push notifications, Customer support, Loyalty programs, Referral incentives	Freemium subscriptions, Bank commissions, Affiliate marketing, In-app advertising, B2B data insights
Cost Structure: Fixed (development, legal, infrastructure setup) + Variable (cloud hosting, marketing, API fees, support)		Competitive Advantage: Real-time AI optimization (unique in India) · Multi-card centralized tracking · Personalized financial intelligence	

Annexure C: Customer Survey Form (Sample Questions)

- Q1. How many credit cards do you currently own? (a) 1 (b) 2–3 (c) 4–5 (d) More than 5
- Q2. Do you actively select which credit card to use based on reward optimization before each transaction? (a) Always (b) Sometimes (c) Rarely (d) Never
- Q3. What is your primary pain point with managing credit card rewards? (Multiple select)
- Q4. Have your reward points ever expired without redemption? (a) Yes (b) No (c) Not Sure
- Q5. Would you use an AI-powered mobile app that recommends the best credit card to use for each transaction? (a) Yes, immediately (b) Yes, after evaluating security (c) Maybe (d) No
- Q6. What is the maximum monthly subscription fee you would pay for such a service?
- Q7. Which feature would you value most? (a) Real-time card recommendation (b) Reward expiry alerts (c) Spending analytics (d) Unified dashboard
- Q8. What is your biggest concern about using such an application?

Do you consent to participate in this survey?

 Copy chart

101 responses

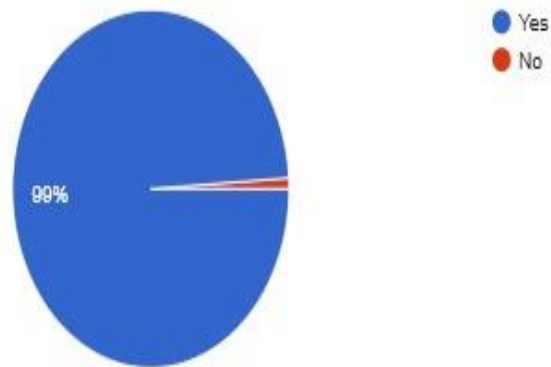


Figure 15: Survey Consent — 99% of Respondents Agreed to Participate

Note: The complete 18-question survey form and 120 filled responses are available as a separate attachment (documentary proof) submitted along with this report.